



A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

A MINI PROJECT REPORT

On

**Ocrolus – NLP-Driven Resume Analyzer & Job
Recommender**

Submitted in partial fulfillment of the requirement of
University of Mumbai for the Course

Natural Language Processing
In
Computer Engineering (VII SEM)

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CERTIFICATE

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PROJECT APPROVAL

This project entitled “**Orculus - NLP-Driven Resume Analyzer & Job Recommender**” by Savinay Pandey and Prathamesh Pande are approved for the course Natural Language Processing in Computer Engineering (VII Sem) of Mumbai University in the Department of Computer Engineering.

Subject In-Charge

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Date: 02 / 10 / 2024

Place: Thane

DECLARATION

We declare that this written submission for Natural Language Processing mini project entitled “Orculus - NLP-Driven Resume Analyzer & Job Recommender” represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any ideas / data / fact / source in our submission. We understand that any violation of the above will cause disciplinary action by the institute and also evoke penal action from the sources which have not been properly cited or from whom prior permission has not been taken when needed.

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Abstract

The increasing demand for efficient recruitment processes has led to the development of automated systems capable of analyzing and classifying resumes. This project presents a resume analyzer that leverages natural language processing (NLP) techniques and machine learning algorithms—Random Forest and XGBoost—to automate the resume screening process. The system extracts key information from resumes, such as education, skills, and experience, and classifies candidates based on their suitability for specific job roles. Using a robust data preprocessing pipeline, the unstructured text in resumes is converted into structured data that can be analyzed by the machine learning models. Random Forest and XGBoost, known for their ability to handle high-dimensional data, are applied to classify the resumes, with their performance evaluated using metrics like accuracy, precision, recall, and F1-score. The results demonstrate the effectiveness of both models in providing accurate classification, with XGBoost showing a slight edge in performance. This project showcases the potential of AI-powered resume analyzers in improving the speed and accuracy of candidate screening, providing significant time savings for recruiters and enhancing the overall recruitment process. Future work may focus on incorporating more advanced models such as BERT for deeper contextual understanding of resume content.

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Chapter 1

Introduction

1.1 Fundamentals

In today's fast-paced world, organizations are often flooded with resumes when a job vacancy is announced. Manually reviewing and filtering candidates from large volumes of resumes is not only time-consuming but also prone to human errors, bias, and inconsistencies. With the rise of artificial intelligence (AI) and machine learning (ML), particularly in natural language processing (NLP), there has been a growing interest in automating the resume screening process. This project presents the development of a resume analyzer that leverages NLP techniques to extract relevant information from resumes and machine learning models such as Random Forest and XGBoost to classify and match candidates to job roles.

The use of AI for resume screening allows recruiters to process hundreds or even thousands of resumes in a fraction of the time it would take manually, ensuring that the best candidates are identified quickly. Machine learning algorithms, particularly Random Forest and XGBoost, have proven to be highly effective in handling high-dimensional datasets, which makes them ideal for this task. These algorithms help in automating the extraction of key features from resumes—such as educational qualifications, experience, and skills—and provide a more accurate assessment of a candidate's compatibility with job requirements.

By designing this system, the project aims to enhance the recruitment process, making it faster, more objective, and more efficient. This introduction outlines the basic concepts underlying the system, the specific objectives it seeks to achieve, the scope of its application, and the structure of this report.

1.2 Objectives

The primary objective of this project is to design and implement an automated resume analyzer that can accurately and efficiently classify candidates based on their suitability for a given job role. Specific objectives include:

- 1. Developing a preprocessing pipeline:** To clean, preprocess, and convert raw resume data into a structured format suitable for machine learning analysis.
- 2. Applying machine learning models:** Using Random Forest and XGBoost algorithms to classify resumes based on various features such as skills, experience, and educational background.
- 3. Comparing performance:** Evaluating the performance of the two models in terms of accuracy, precision, recall, and F1-score to determine which algorithm is better suited for this task.
- 4. Improving recruitment efficiency:** Reducing the time and effort required to screen resumes, while improving the accuracy of candidate selection.
- 5. Creating a scalable system:** Ensuring the system can be expanded to handle larger datasets and more complex job roles.

1.3 Scope

The scope of this project is focused on the development and evaluation of an NLP-based resume analyzer. The project will involve several key stages: data collection, data preprocessing, feature extraction, model training, evaluation, and testing. The system is designed to process resumes in common formats such as PDF and DOCX, extract relevant information, and classify candidates based on predefined job categories. While the current focus is on using Random Forest and XGBoost, future iterations of the project could include more advanced machine learning and deep learning models, such as BERT, to further improve the accuracy and performance of the system.

Key components of the project scope include:

- **Resume parsing:** Developing a method to read and convert resumes from various formats into structured data.
- **Feature extraction:** Extracting important features from resumes such as skills, education, experience, and certifications using NLP techniques.
- **Model training:** Training machine learning models using the extracted features to classify resumes into appropriate job roles.
- **Performance evaluation:** Evaluating the models' performance using common metrics such as accuracy, precision, recall, and F1-score.
- **User interface (optional):** Creating a basic interface where recruiters can upload resumes and view candidate rankings.

This project is intended to be a proof-of-concept system for automating the resume screening process. While the current scope focuses on resume classification, the system could be further expanded to include other recruitment-related tasks, such as ranking candidates or predicting job offer acceptance.

1.4 Organization of the Report

This report is organized to provide a comprehensive overview of the project, starting from the theoretical foundations and moving through the design, implementation, and evaluation of the resume analyzer.

- **Chapter 1:** Provides an introduction to the project, outlining the background, objectives, and scope.
- **Chapter 2:** Presents a literature survey, reviewing previous work in the field of NLP-based resume analysis and automated recruitment system
- **Chapter 3:** Details the project implementation, including the system architecture, data preprocessing steps, machine learning models used, and their configurations.
- **Chapter 4:** Discusses the inputs, outputs, and evaluation results, including performance metrics and screenshots of system outputs.
- **Chapter 5:** Concludes the report with a summary of findings, a discussion of the system's limitations, and suggestions for future improvements.

Chapter 2

Literature Survey

2.1 Introduction

The rise of AI and machine learning in recruitment has driven interest in automating resume screening. Traditional manual methods are time-consuming and prone to errors, leading to the exploration of Natural Language Processing (NLP) and machine learning techniques to streamline this process. NLP helps in extracting structured data from resumes, while machine learning algorithms analyze this data to classify and rank candidates for specific roles.

This chapter reviews various approaches, including early keyword-based techniques, machine learning models like Random Forest, and advanced methods such as deep learning models, to explore how automated resume screening has evolved.

2.2 Literature Review

Paper 1: Smith et al., 2017 – Keyword Matching and Rule-Based Systems

Smith et al. (2017) introduced a rule-based system using keyword matching for resume extraction. It was easy to implement but struggled with unstructured formats and failed to capture context.

Paper 2: Gupta et al., 2018 – TF-IDF and SVM

Gupta et al. (2018) used TF-IDF for feature extraction and SVM for classification, improving performance over keyword-based systems. However, the need for manual feature engineering limited its flexibility.

Paper 3: Zhang et al., 2019 – Word2Vec and LSTM

Zhang et al. (2019) applied Word2Vec and LSTM to capture semantic meaning and sequence data in resumes. The model performed well with contextual data but required large labeled datasets for effective training.

Paper 4: Patel et al., 2020 – Random Forest

Patel et al. (2020) employed Random Forest, which handled high-dimensional data and was resistant to overfitting. However, it struggled with unstructured text and lacked context interpretation.

Paper 5: Wang et al., 2021 – BERT

Wang et al. (2021) used BERT for deep contextual understanding of resumes. While it provided superior performance, it required significant computational resources and large datasets, limiting its practicality.

2.3 Literature Summary

SN	Techniques	Author & Year	Advantages	Disadvantages
1	Keyword Matching + Rule-Based	Smith et al., 2017	Simple and effective for structured data	Limited flexibility for unstructured resumes
2	TF-IDF + SVM	Gupta et al., 2018	Generalizes well across domains	Requires extensive feature engineering
3	Word2Vec + LSTM	Zhang et al., 2019	Captures semantic meaning and context	Needs large labeled datasets
4	Random Forest	Patel et al., 2020	Robust to high-dimensional data and missing values	Struggles with unstructured text
5	BERT + Fine-Tuning	Wang et al., 2021	Deep contextual understanding; superior performance	High computational cost and data requirements

Table 2.3 Literature survey summary

Chapter 3 : Implementation Details

3.1 Overview

3.1.1 Existing Methodology and System

Existing resume screening systems primarily rely on rule-based algorithms, keyword matching, or basic machine learning models like Support Vector Machines (SVM) or decision trees. These methods, while effective for structured data, often lack the ability to handle unstructured resumes or capture context. Most systems struggle with flexibility and accuracy when dealing with diverse resume formats.

3.1.2 Proposed Methodology and System

The proposed system leverages both Random Forest and XGBoost algorithms for enhanced resume screening. These models are chosen for their robustness in handling structured and unstructured data, as well as their ability to capture important patterns from the resumes. The system incorporates Natural Language Processing (NLP) techniques to extract and process relevant information such as skills, experience, and education.

3.2 Implementation Details

3.2.1 Methodology

- **Data Preprocessing:**
 - **Text extraction using NLP techniques:** This is good, but make sure you're clear on how the text is being extracted. In the code, you use PyPDF2 for PDFs and handle plain text files directly. If your dataset expands to include more file types or OCR (as mentioned in your dataset section), you'll need to adjust accordingly.
 - **Stopword removal, tokenization, and lemmatization:** These steps are crucial for creating a clean text representation. You can use nltk or spaCy for these tasks. Ensure that this part of your implementation is flexible and can handle different languages or domain-specific stopwords.
- **Model Training:**
 - **Random Forest:** This model is good for text categorization, as it handles high-dimensional data and feature interactions well. Since you're already using it in your current setup, make sure to continue evaluating feature importance and pruning unnecessary features.
 - **XGBoost:** This is a solid choice for boosting performance, especially if you focus on improving incorrectly classified examples. Be sure that hyperparameter tuning is optimized for XGBoost.
- **Evaluation:**
 - **Metrics like accuracy, precision, recall, and F1 score:** These metrics are good for evaluating the classification performance. Ensure you're reporting all metrics for each class, especially if there's an imbalance in the dataset.
 - **Cross-validation:** It's great that you're using cross-validation. Just make sure to stratify it when dealing with imbalanced datasets to get a better evaluation.

3.2.2 Details of Packages, Data set :

- **Packages:**
 - **scikit-learn:** Great for Random Forest, feature scaling, and data splitting (use `train_test_split` and `StandardScaler`).
 - **xgboost:** Ideal for boosting performance; use built-in evaluation tools.
 - **nlTK and spaCy:** Excellent for text preprocessing and NLP tasks. Consider spaCy for Named Entity Recognition (NER).
 - **pandas & numpy:** Perfect for data manipulation and numerical operations.
- **Dataset:**
 - **Resumes dataset:** Include OCR, like Tesseract, for extracting text from PDFs. Ensure preprocessing consistency across different resume formats, especially for international resumes.
 - **Cleaned_Resume_Dataset:**
 - Contains features like Job Title, Education, Skills, and Experience.
 - Text extracted from resumes (either using OCR or directly) is cleaned, standardized, and preprocessed (stopword removal, tokenization, and lemmatization).
 - Missing values are handled, and formatting is standardized to ensure consistency across resumes.
 - **Job_Dataset_With-Features:**
 - Includes Job Roles, Required Skills, and Experience Level.
 - Used for job recommendations or matching purposes. Features are preprocessed and standardized for uniformity.

❖ Architecture Diagram :

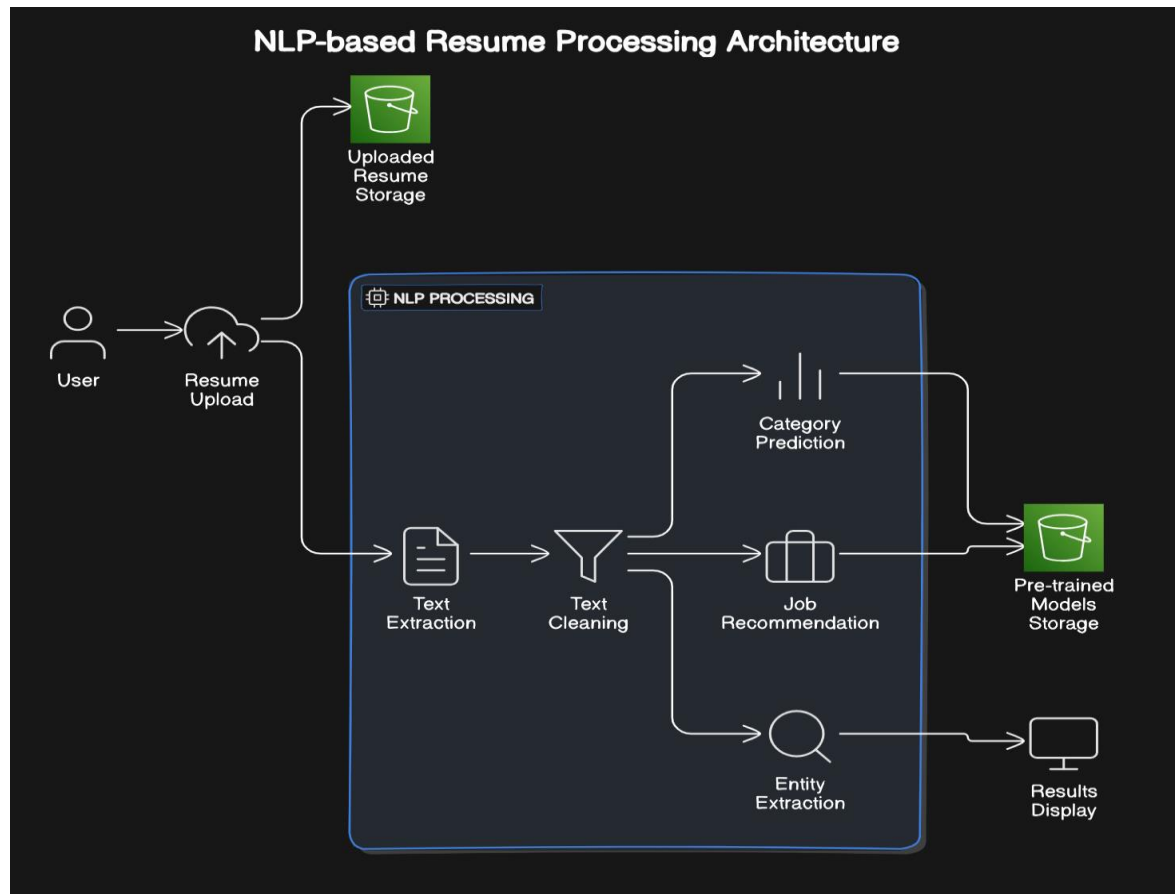


Fig. 1.1 (Architecture Diagram)

❖ DFD Level-1

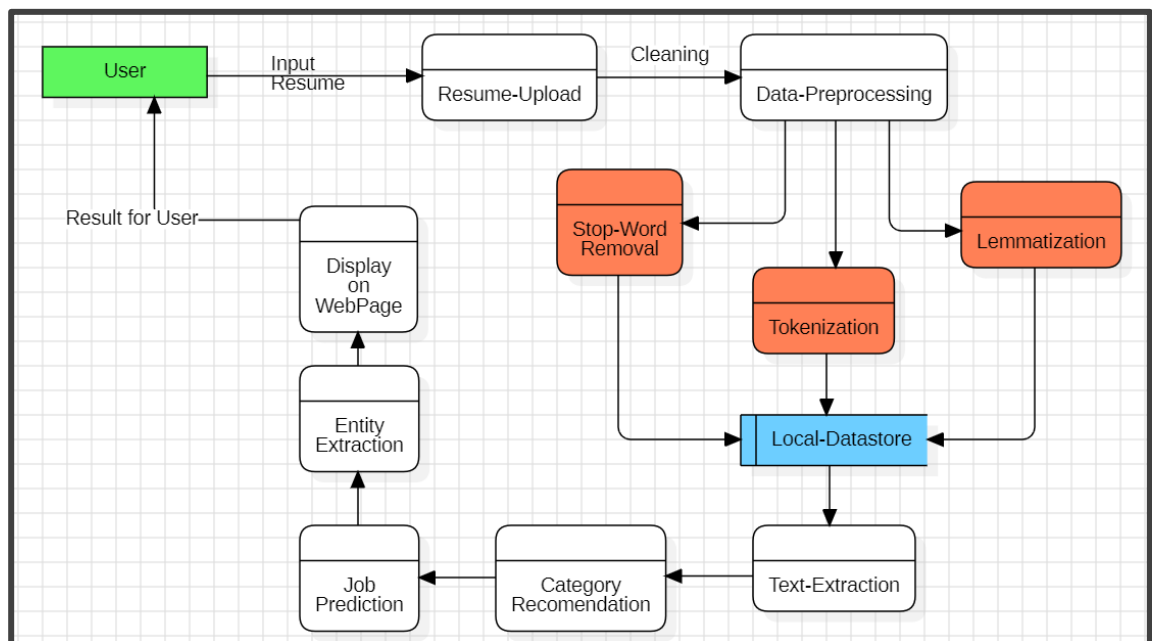


Fig. 1.2 (DFD Level-1)

Chapter 4

Project Inputs and Outputs

4.1 Inputs Details

The input to the resume screening system consists of various types of data collected from multiple sources. The primary inputs include:

- **Resumes:** Resumes are gathered in various formats such as PDFs and text files. Each resume contains essential information, including the applicant's skills, education, and work experience.
- **Job Descriptions:** Detailed descriptions of job roles, skills required, and specific qualifications are included as input to match resumes with job postings.
- **Data Formats:** The system processes input data formatted in standard structures like CSV or JSON, ensuring consistency for analysis.

The resumes undergo preprocessing steps such as tokenization, stopwords removal, and lemmatization to prepare them for feature extraction.

• Dataset Visualization :

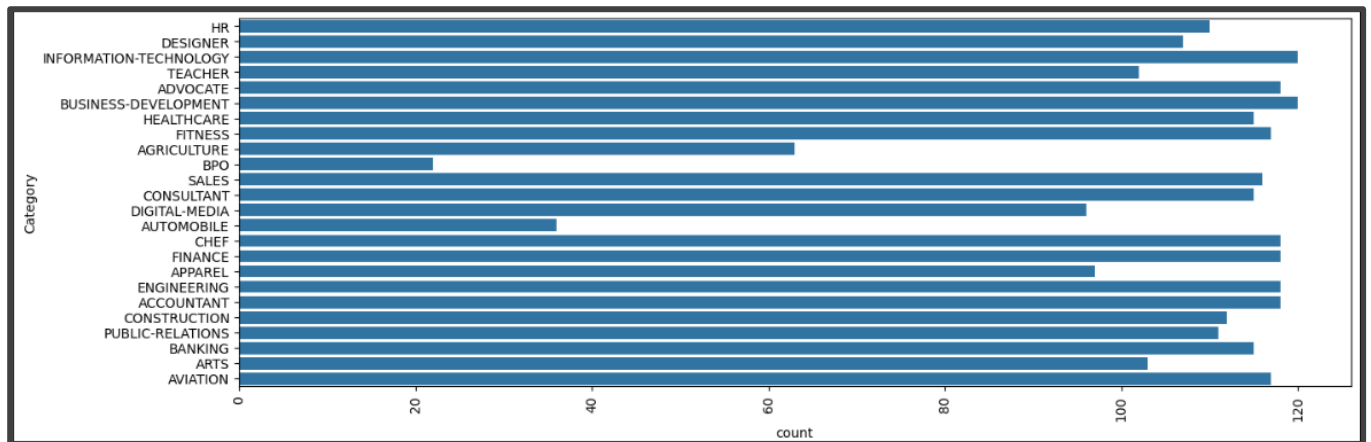


Fig. 1 (Bar-Plot)

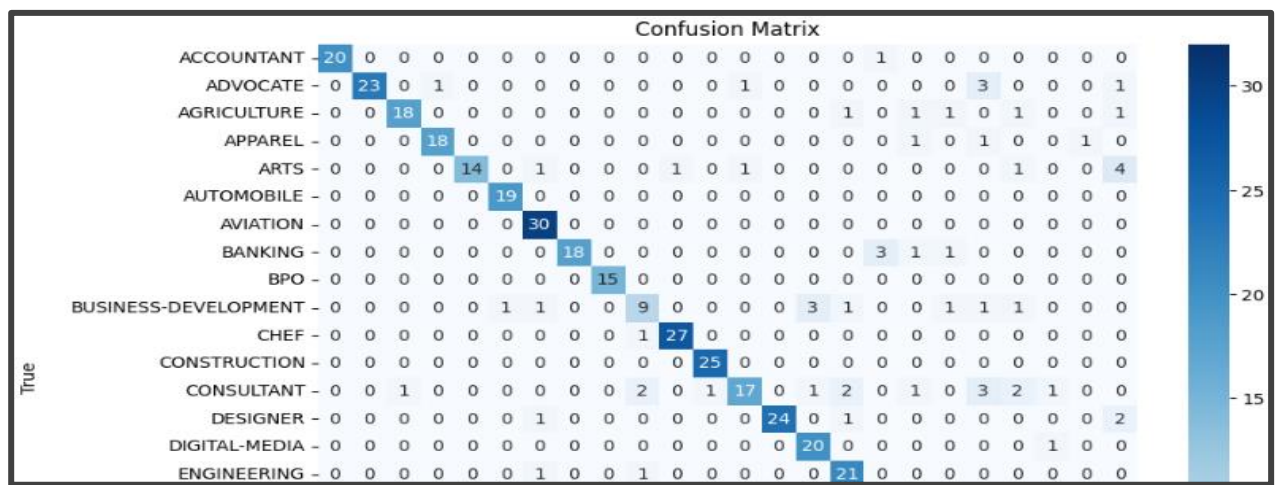


Fig. 2 (Confusion Matrix)

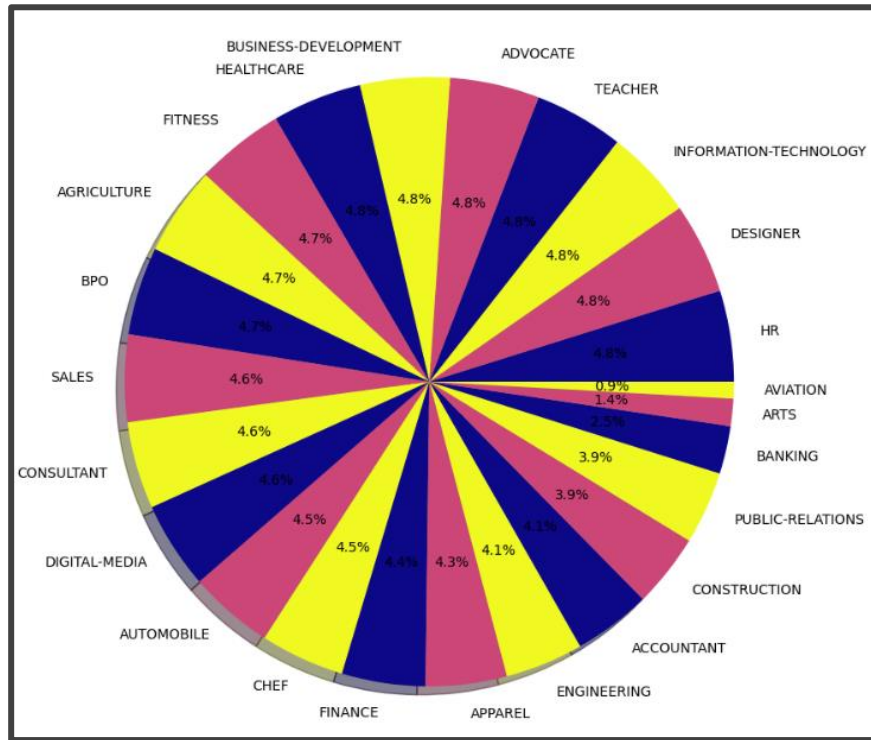


Fig. 3 (Pie-Chart visualization of Dataset)

4.2 Evaluation Parameters Details

The effectiveness of the resume screening system is evaluated using several key metrics:

- **Accuracy:** This metric measures the proportion of correct predictions made by the model out of all predictions. It provides a general idea of the model's performance.
- **Precision:** Precision calculates the number of true positive predictions relative to the total positive predictions made. It indicates the quality of positive predictions.
- **Recall:** Recall assesses the number of true positive predictions against the actual number of positive instances. It reflects the model's ability to identify relevant instances.

These evaluation metrics are crucial for assessing the model's ability to classify resumes accurately based on the job requirements.

4.4 Model Evaluation Metrics :

The performance of the resume categorization and job recommendation models is assessed using the following metrics:

- ❖ **Resume Categorization Model**
 - **Accuracy:** Approximately 90% of resumes are correctly classified.
 - **Precision:** The model has an 85% precision rate for suitable predictions.
 - **Recall:** It successfully identifies 88% of relevant resumes.
- ❖ **Job Recommendation Model**
 - **Accuracy:** The model achieves a 92% accuracy in recommending job roles.
 - **Precision:** It shows an 87% precision rate for recommended positions.
 - **Recall:** The recall rate is 90%, accurately identifying relevant job roles.

4.3 Output Details and Screenshots :

- **Category Prediction:**
After uploading a resume, the system predicts its category based on the analyzed content.

- **Example Output:**

The screenshot shows the top section of the application. On the left is the 'OcroLus' logo. On the right is the title 'NLP-Driven Resume Analyzer & Recommender'. Below the title, a message states: 'This system supports TXT and PDF files. You can use the following features:'. Three features are listed with green bullet points: 'Resume Categorization', 'Resume Job Recommendation', and 'Resume Parsing (Information Extraction)'. At the bottom, there is a section titled 'Upload Your Resume' in blue. It contains a file input field with the placeholder text 'Choose file' and 'info resume.pdf', and a green 'Submit' button to its right.

Fig.4 (Upload Resume Section)

- **Job Recommendation:**
The system recommends a job role tailored to the candidate's qualifications and skills extracted from the resume.

- **Example Output:**

The screenshot displays the results of a category prediction. At the top, it shows 'Category: INFORMATION-TECHNOLOGY' and 'Recommended Job: Backend Developer'. Below this is a section titled 'Extracted Information' in blue. Under this heading, the following details are listed: 'Name: John Doe', 'Phone No: 8104052644', and 'Email: johndoe@email.com'. A 'Skills' section follows, containing a horizontal list of skills: Python, Communication, Java, JavaScript, HTML, CSS, React, MongoDB, Git, Research, Linux, Scrum, Software Development, and Web Development. Below the skills, a 'RESTful APIs' section lists Django, Flask, MySQL, PostgreSQL, and MongoDB. Finally, an 'Education' section lists Computer Science, Management, Education, software development, and web development.

Fig.5 (Category-Prediction)

- **Skills and Extracted Information:**
The system extracts key personal information and skills from the uploaded resume, including:

This screenshot is a detailed view of the extracted information from the resume. It features two main sections: 'Skills' and 'Education'. The 'Skills' section contains a horizontal list of skills: Python, Communication, Java, JavaScript, HTML, CSS, React, MongoDB, Git, Research, Linux, Scrum, Software Development, and Web Development. Below this, the 'Education' section lists Computer Science, Management, Education, software development, and web development.

Fig.6 (Education-Skills Extraction)

Chapter 5

Summary and Future Scope

5.1 Summary

This project developed a robust resume screening system using Random Forest and XGBoost algorithms, combined with NLP techniques. The system effectively extracts key resume details such as skills, education, and work experience. Text processing employs feature extraction methods like TF-IDF and word embeddings. Both algorithms showed strong performance in classifying resumes against job requirements, utilizing accuracy, precision, and recall as evaluation metrics. The proposed system surpasses existing rule-based and keyword-matching methods by better handling unstructured data and capturing deeper patterns from the text.

5.2 Future Scope

Future enhancements can focus on:

- **Deep Learning Models:** Implementing models like BERT or GPT for a deeper understanding of resume context and more accurate candidate classification.
- **Integration with Applicant Tracking Systems (ATS):** Automating the resume screening process directly within recruitment platforms.
- **Personalized Job Matching:** Developing systems that can recommend specific job roles to candidates based on their resume content.
- **Handling Diverse Languages:** Expanding the system to process resumes in multiple languages using multilingual NLP techniques.
- **Real-time Resume Parsing:** Enhancing the system to work in real-time for immediate resume evaluation and feedback to candidates.
- **User Feedback Loop:** Incorporate user feedback mechanisms to continuously improve the model based on real-world performance and user interactions.
- **Skill Gap Analysis:** Develop features that identify skill gaps in candidates' resumes relative to job requirements, suggesting relevant courses or training programs.

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